

QX4001 – Molecules to Materials

Rotation 1 – Question Set 2

- 1) What is the oxidation state of Fe in the following compounds: FeO and Fe₂O₃.
- 2) Explain why the most common oxidation states of Fe are Fe(II) and Fe(III) (leading to ions Fe²⁺ or Fe³⁺).
- 3) Non-metals tend not to be as shiny as metals. There are exceptions. Can you name at least one?
- 4) Explain the trend in atomic radius as you go from halogens to their closest alkaline metals in atomic number (for example from F to Na)
- 5) Explain the trend in ionic radius as you go from halogens to their closest alkaline metals in atomic number (for example from F⁻ to Na⁺)
- 6) Explain the trend in atomic radius as you go from alkaline to earth alkaline metals (for example from Li to Be, or K to Ca)
- 7) Explain the trend in 1st ionisation energy within the main group as you go left in the periodic table (for example from Li all the way to F and Ne)
- 8) Explain the trend in 1st ionisation energy within the main group as you go down in the periodic table in the same group (for example from Li all the way to Cs)
- 9) Explain the trend of ratios (Nth ionisation energy / 1st ionisation energy) where N is 2 (2nd ion. en.), 3 (3rd ion. en.), 4 (4th ion. en.), etc, as you go right on the periodic table within the same period. (for example from Li to F, Na to Cl, etc).
- 10) Explain the trend in electronegativity as you go to the right of the periodic table within the main group (e.g. from Li to F) and above within the same group (e.g. from Cs to Li or from I to F).